

Speech & Audio

Representations, understanding and generation

Yunyun Wang

misc.lyra@gmail.com

What can speech & audio models do?

Two people talking in a noisy café



Automatic speech recognition (ASR)

What was said?

Enhancement

Can we hear it clearly?

Speaker diarization

Who spoke when?

Voice conversion

Can we change the voice?

Source separation

Which sound source?

Generation

Can we create new speech?

What's in a recording?

01

Content

What was said

02

Speaker

Who it sounds like

03

Delivery

How it was said

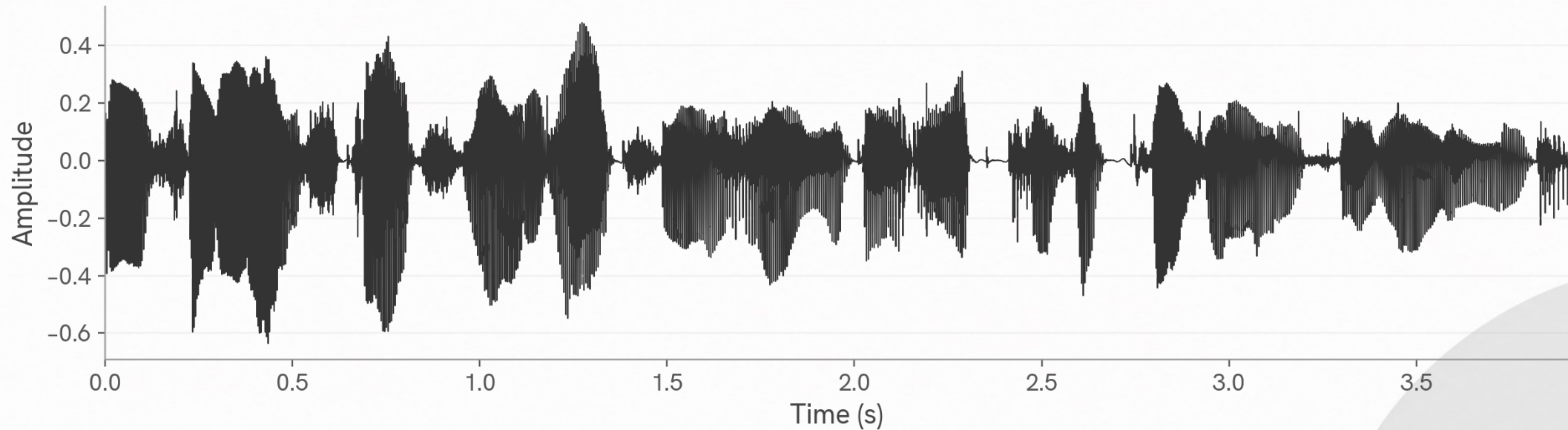
04

Environment

Where it was recorded

A waveform is a measurement

Real speech recording, displayed as amplitude over time

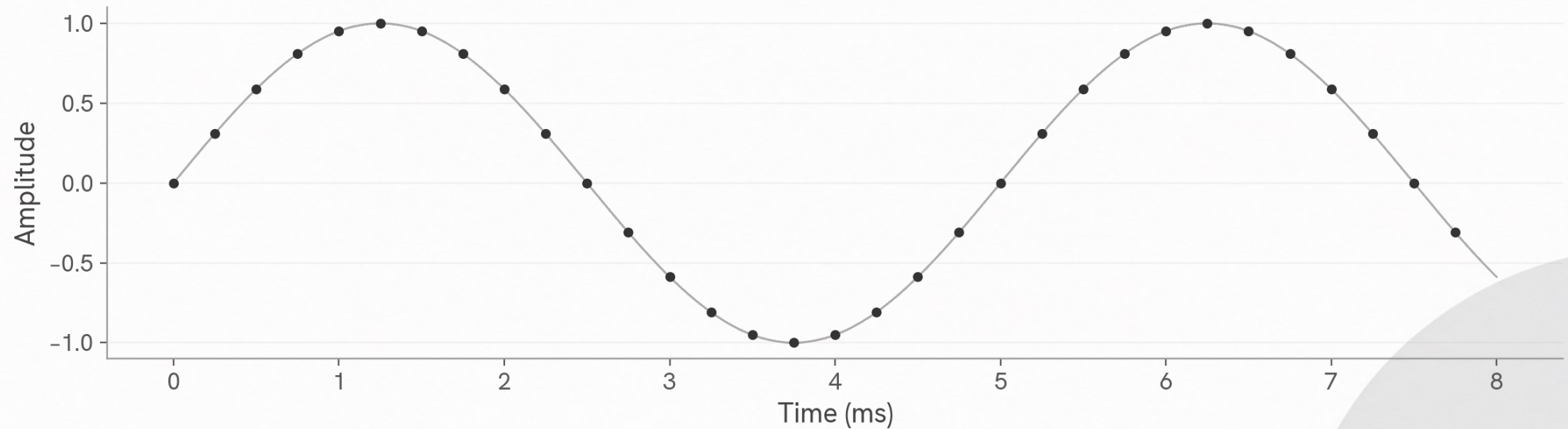


Play the recording

3.9 s

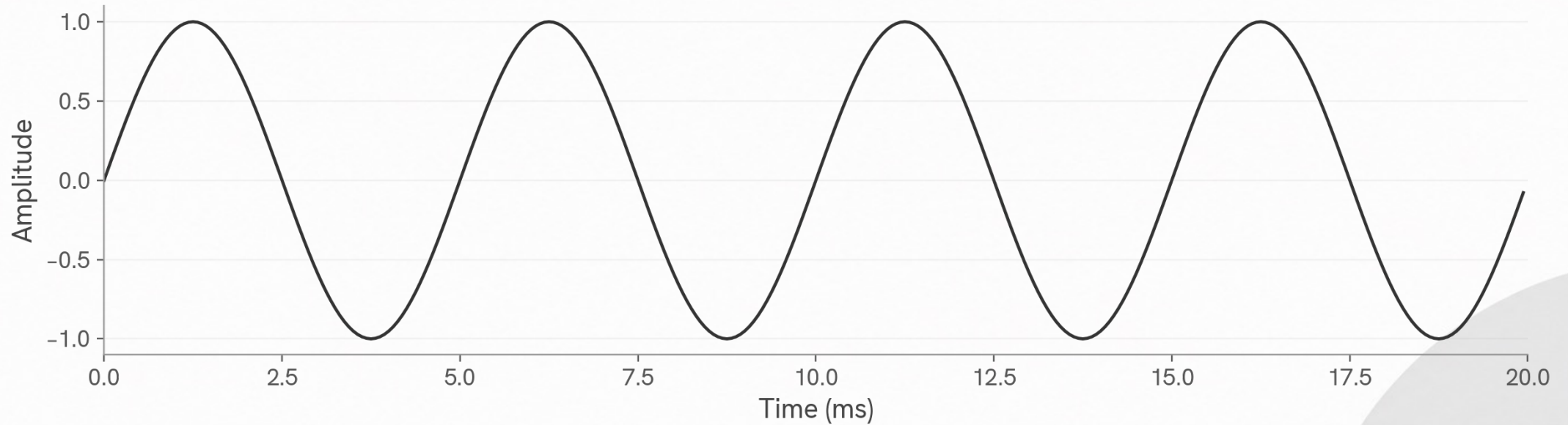
Sampling makes the signal discrete

A synthetic tone sampled at regular intervals



200 Hz

200 cycles per second

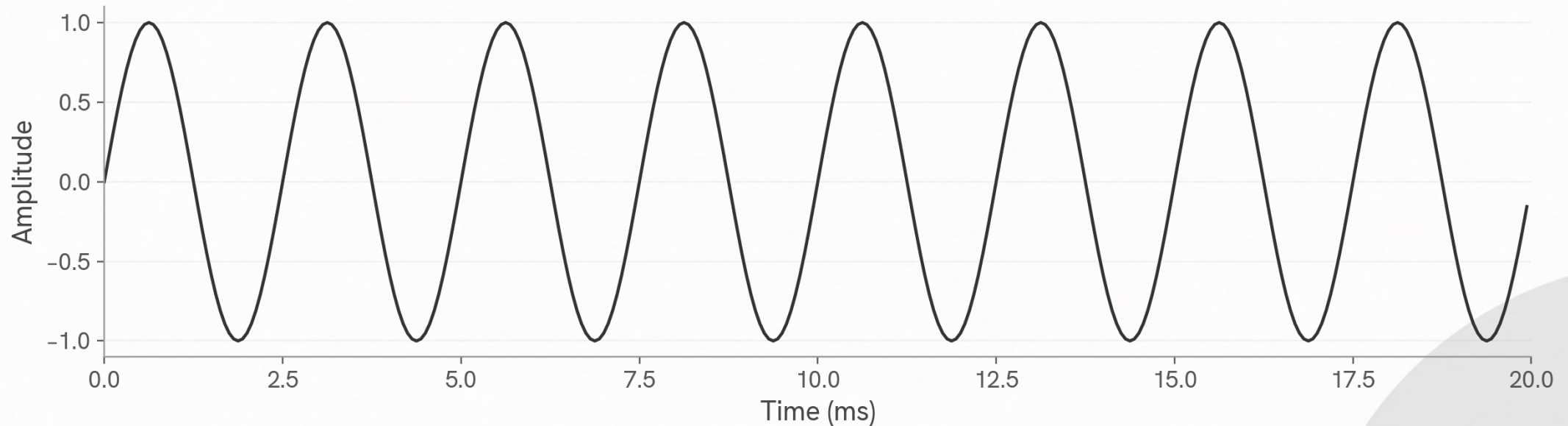


Play tone

2.0 s

Frequency is physical. Pitch is perceived

Double the frequency. One octave higher.



Play tone

2.0 s

Same fundamental. Different sound

Listen to A, then B.

**A**

2.0 s

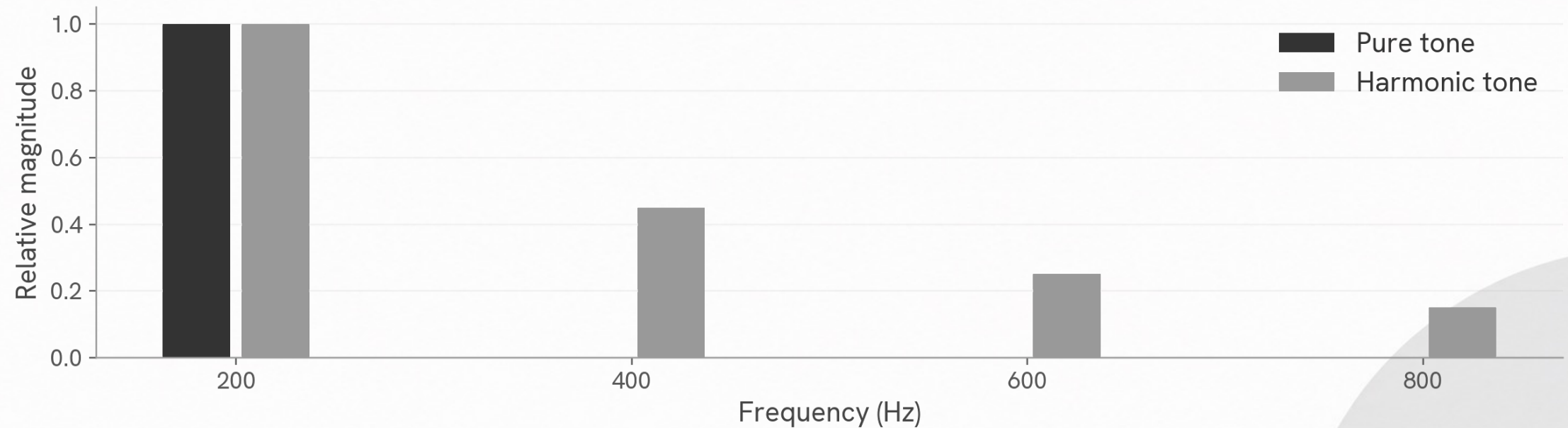
**B**

2.0 s

Listen first. Then compare.

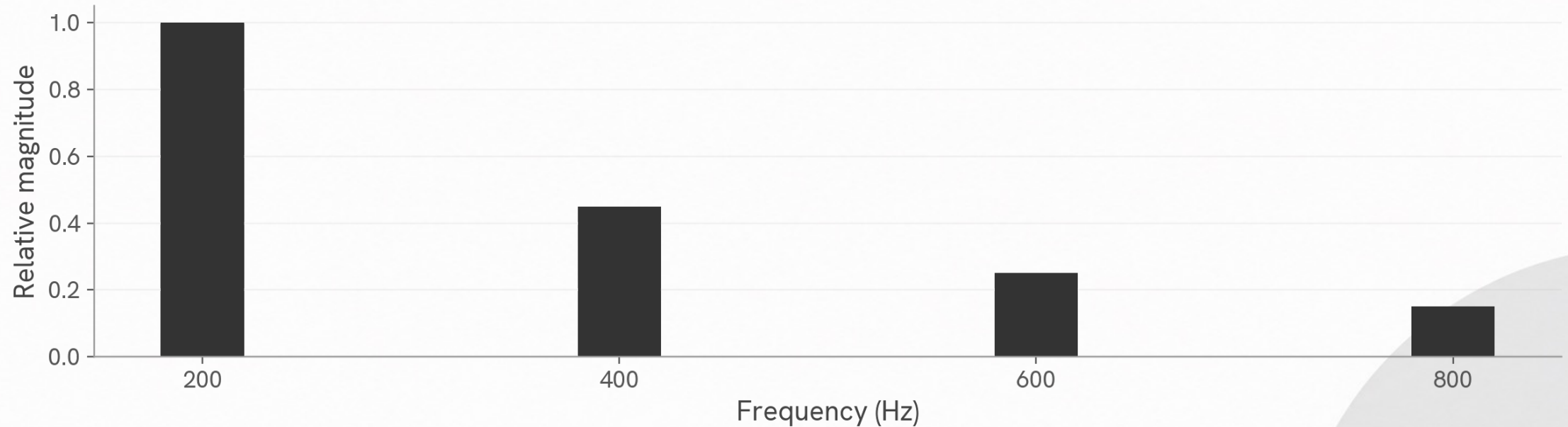
Harmonics shape timbre

200 Hz fundamental with energy at its multiples



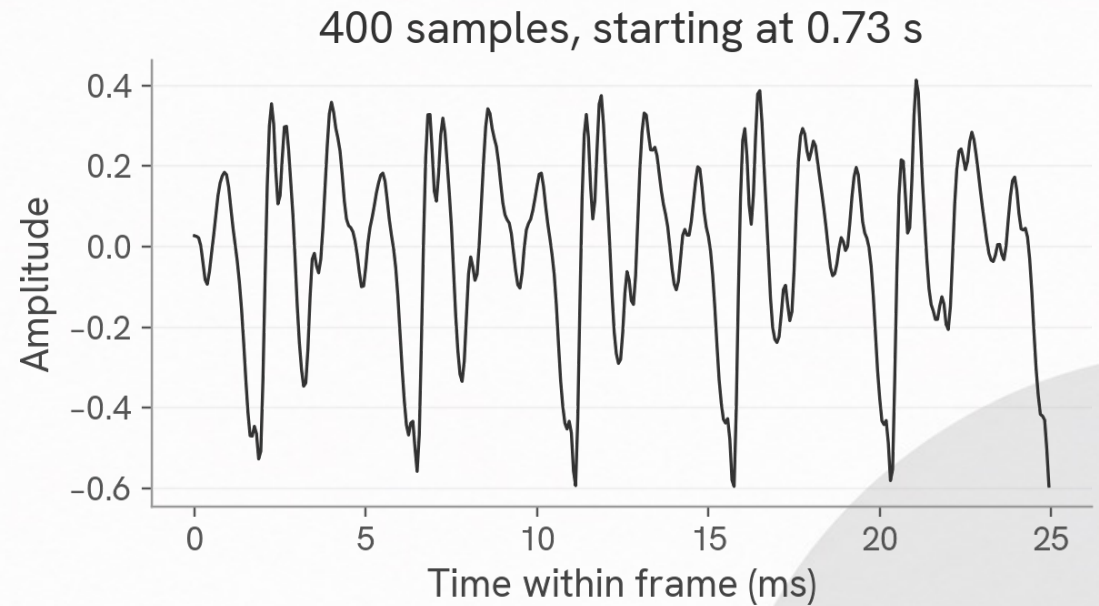
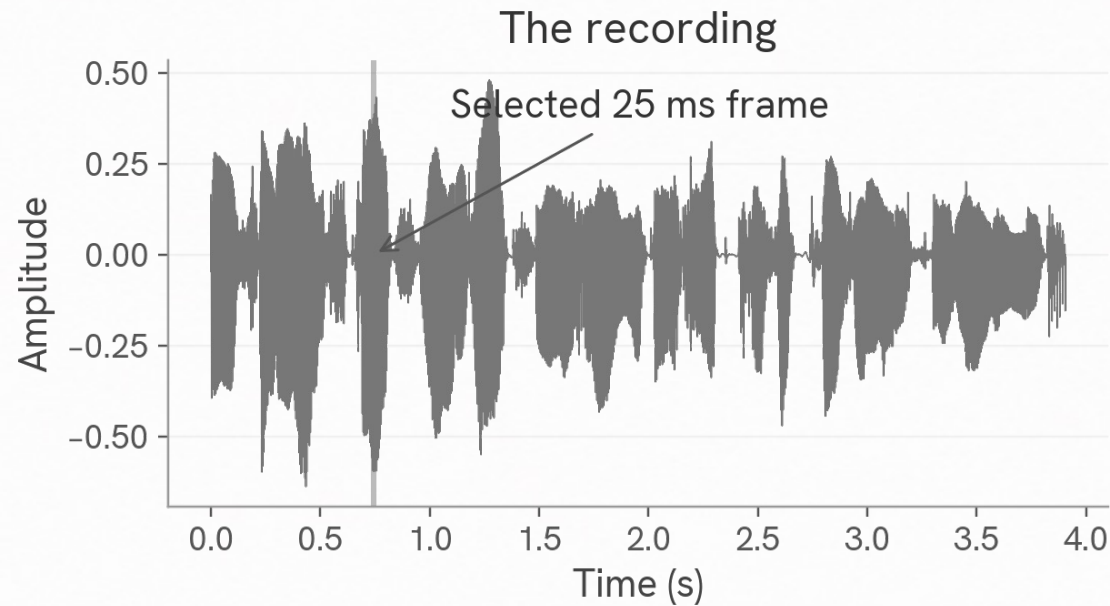
Reading a spectrogram column

Frequency components of one frame, shown as a bar plot



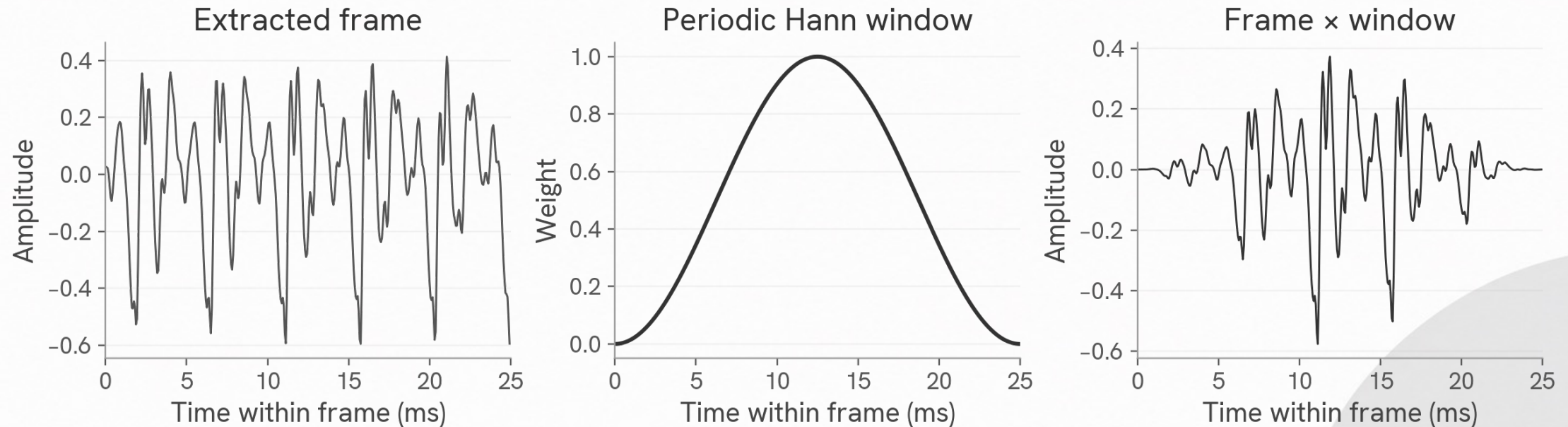
Short-Time Fourier Transform (STFT)

One 25 ms segment contains 400 samples at 16 kHz



The Hann window weights each sample

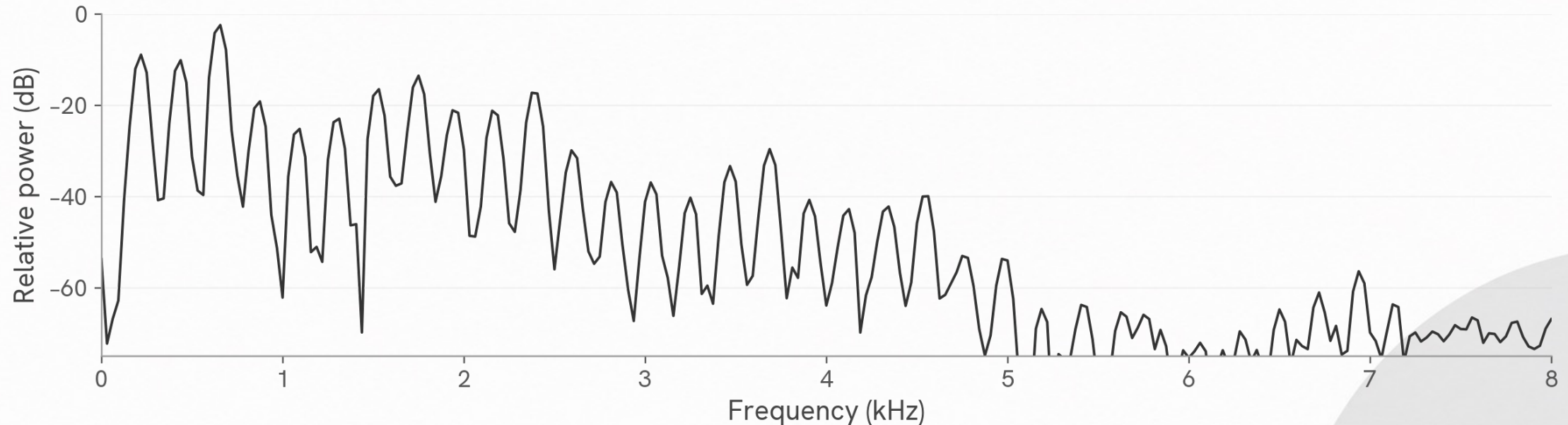
Apply a smooth taper before the Fast Fourier Transform (FFT)



$$x_m[n] = x[mH + n] w[n] \quad \cdot \quad n = 0, \dots, 399$$

The FFT produces frequency coefficients

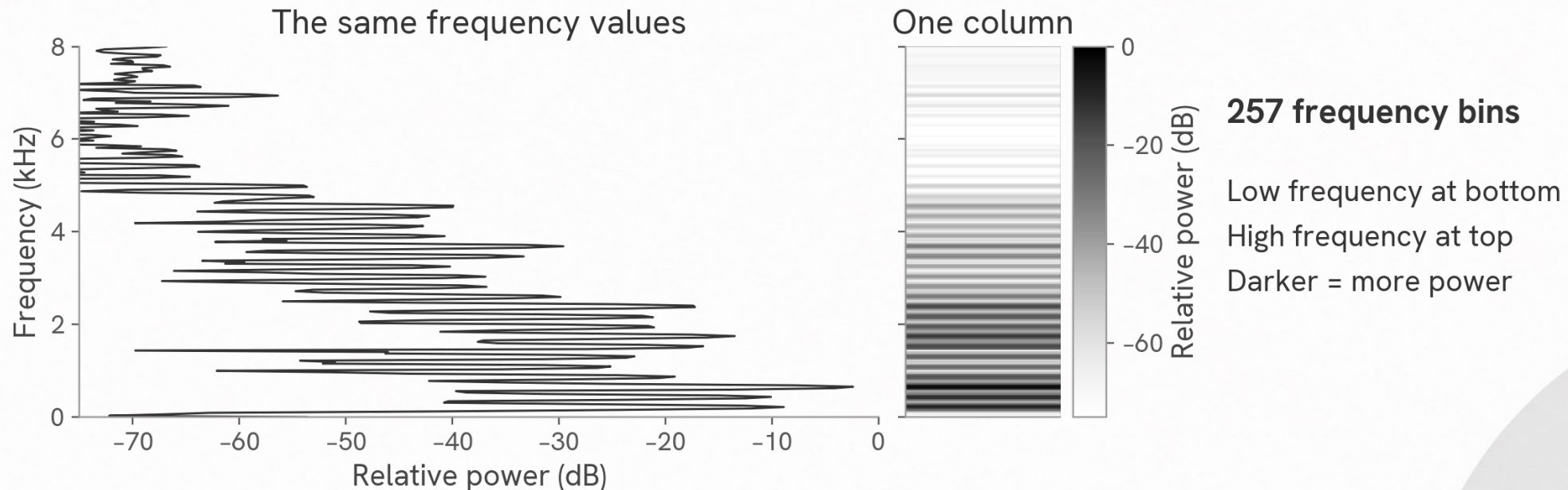
400 windowed samples, zero-padded to a 512-point FFT



$$X_m[k] = \text{FFT}(x_m)[k] \quad \cdot \quad P_m[k] = |X_m[k]|^2 \quad \cdot \quad f_k = k \times 31.25 \text{ Hz}$$

Building one spectrogram column

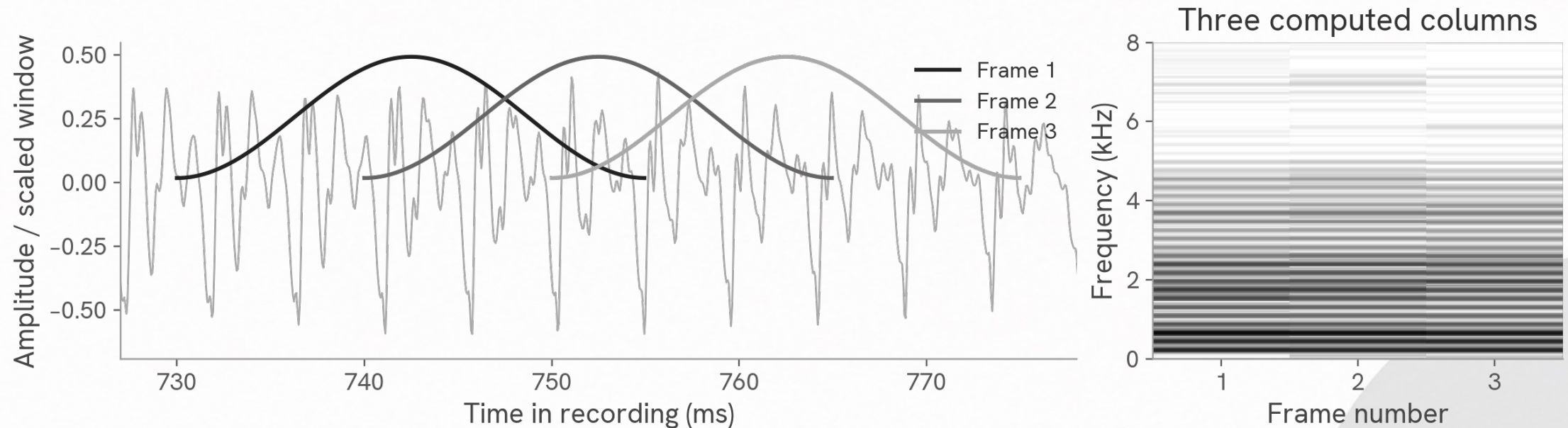
The curve and the strip show exactly the same 257 values



Column at $t = 0.7425$ s · frequency determines height; power determines shade

The hop moves the window forward

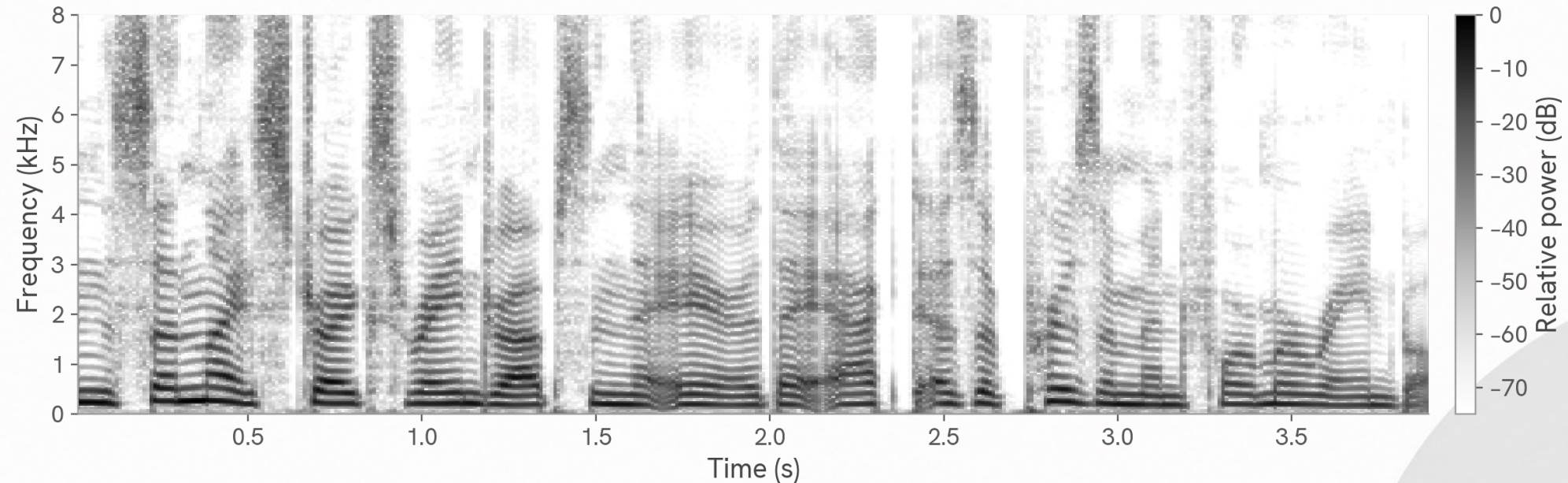
A 10 ms hop advances by 160 samples



Window = 25 ms · hop = 10 ms · overlap = 15 ms

The spectrogram stacks all those columns

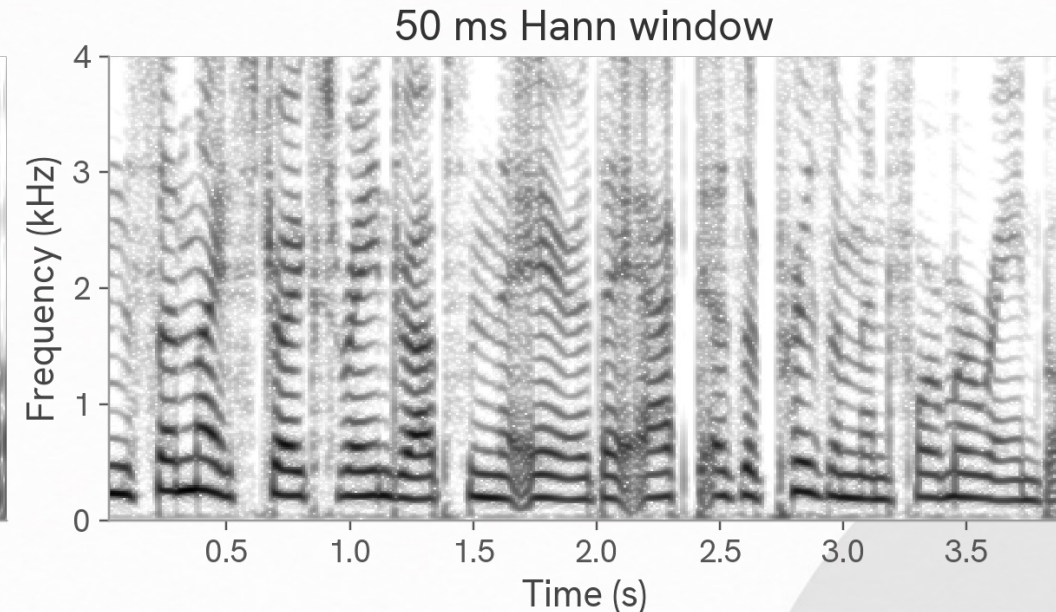
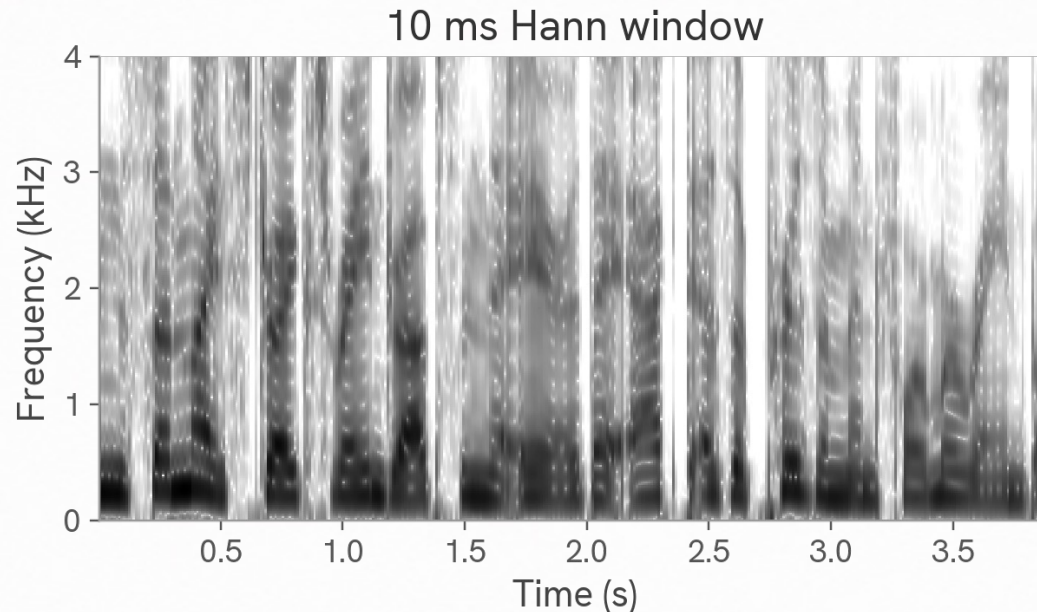
Time on the horizontal axis, frequency on the vertical axis



16 kHz audio · 25 ms Hann · 10 ms hop · 512-point FFT

Window length changes what we can resolve

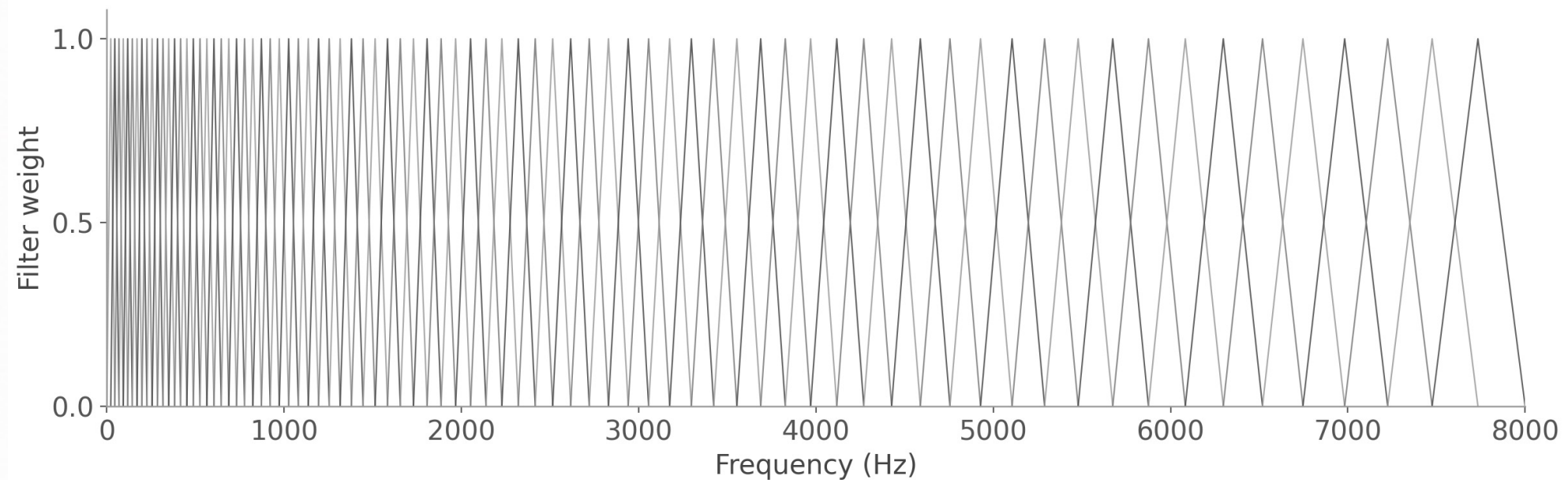
The same recording with 10 ms and 50 ms Hann windows



Longer window: finer frequency detail, coarser time localization

Mel filters pool neighboring frequency bins

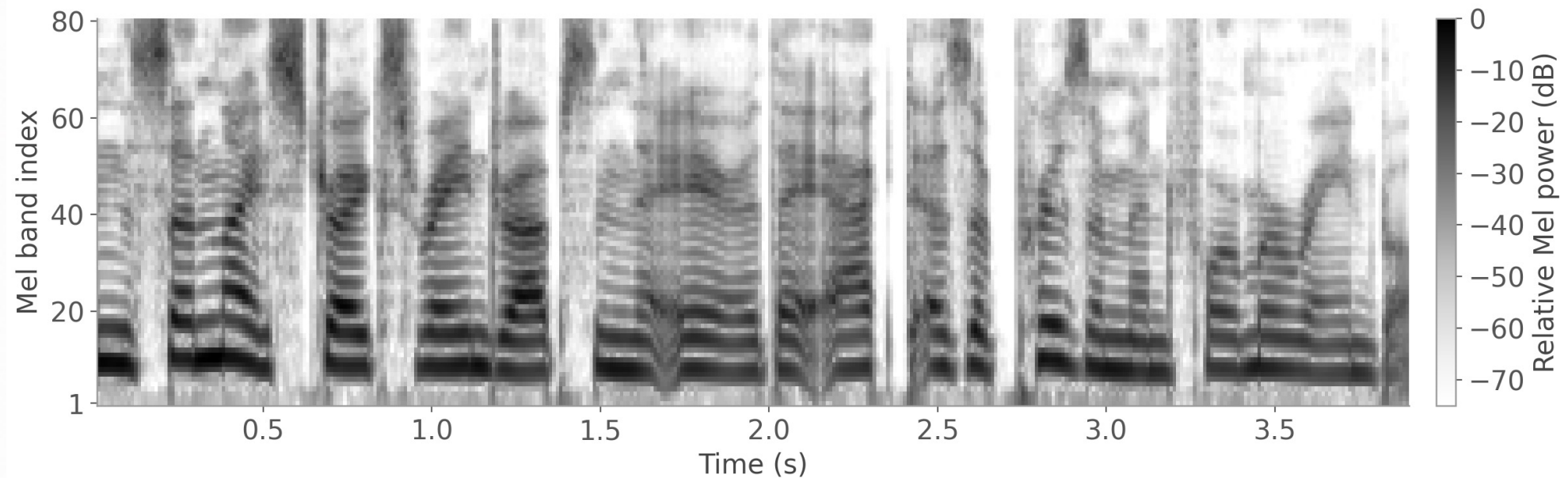
80 overlapping triangular filters on the Mel scale



$$M_m[b] = \sum_k F[b,k] P_m[k] \quad \cdot \quad \text{pool power first, then take the logarithm}$$

The Mel spectrogram uses the same frames

80 Mel bands per frame, computed from the real speech recording



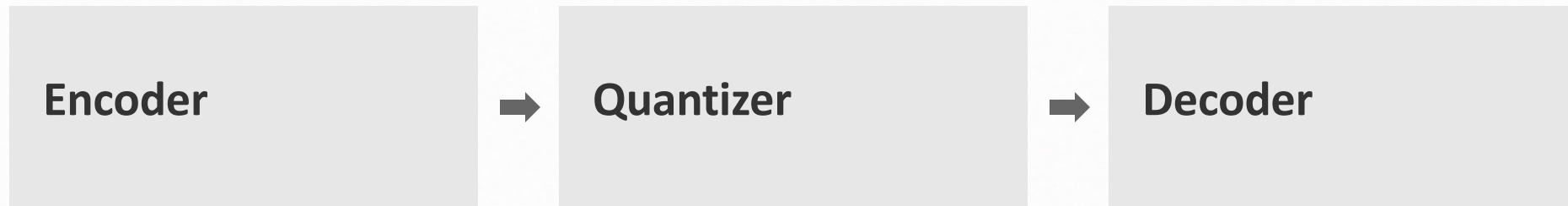
Same time frames · 257 FFT bins become 80 Mel bands

Audio becomes a long sequence

16,000–48,000 samples per second, per channel
48 kHz is common for high-fidelity recording.

A neural audio codec

Waveform in. Reconstructed waveform out.



Residual Vector Quantization (RVQ)

Each stage subtracts a contribution and adds it to the reconstruction

$$\mathbf{r}_0 = \mathbf{z}$$



Quantizer 1



$\mathbf{q}_1$

$$\mathbf{r}_1 = \mathbf{z} - \mathbf{q}_1$$



Quantizer 2



$\mathbf{q}_2$

$$\mathbf{r}_2 = \mathbf{z} - \mathbf{q}_1 - \mathbf{q}_2$$



Quantizer 3



$\mathbf{q}_3$

$$\mathbf{r}_i = \mathbf{r}_{i-1} - \mathbf{q}_i$$

$$\text{Reconstruction} = \mathbf{q}_1 + \mathbf{q}_2 + \mathbf{q}_3$$

Inside the Descript Audio Codec (DAC)

Nearest-neighbor lookup happens in a projected space



Lookup compares unit-length vectors (Euclidean / L2 normalization).

Retrieve the original entry, then project it back before subtracting and summing.

The decoder adds vectors, not token IDs

Each stage transmits an index into its own codebook

Index c_1 $\rightarrow$ Codebook 1 + output projection $\rightarrow q_1$

Index c_2 $\rightarrow$ Codebook 2 + output projection $\rightarrow q_2$

Index c_3 $\rightarrow$ Codebook 3 + output projection $\rightarrow q_3$

$q_1 + q_2 + q_3$ $\rightarrow$ Waveform decoder

Acoustic detail and linguistic structure

Different training objectives emphasize different information.

Acoustic representations

Reconstruct the recording

Semantic-oriented representations

Capture useful speech structure

wav2vec 2.0: predict what was masked

Learn speech structure from unlabeled recordings

Waveform → latent frames

Quantized targets

Mask selected positions

Contrastive candidates

Context network predicts

Useful contextual features

Hidden-Unit BERT (HuBERT)

Bidirectional Encoder Representations from Transformers (BERT)
Offline clustering supplies the masked-prediction targets.

Acoustic features

Offline clustering

Pseudo-label sequence

Mask the input

Predict cluster labels

Refine representations

The representation sets the trade-off

Reconstruction: preserve detail

Understanding: expose structure

Generation: balance both

ASR recovers the words

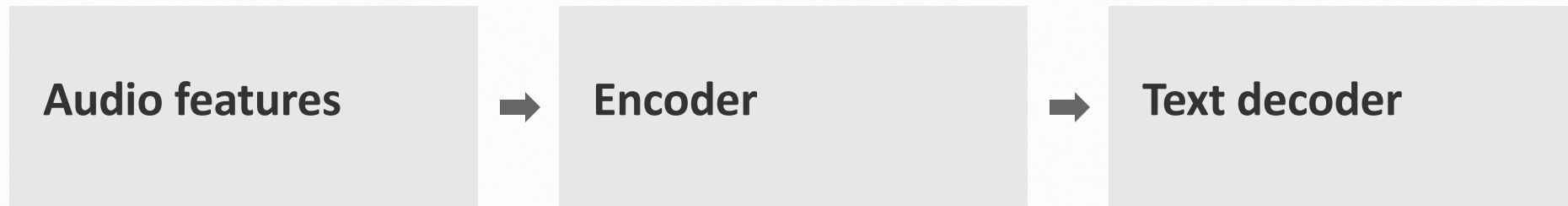
Audio → words

Café recording → **“Can you hear me?”**

Word Error Rate (WER)
= (substitutions + deletions + insertions) / reference words

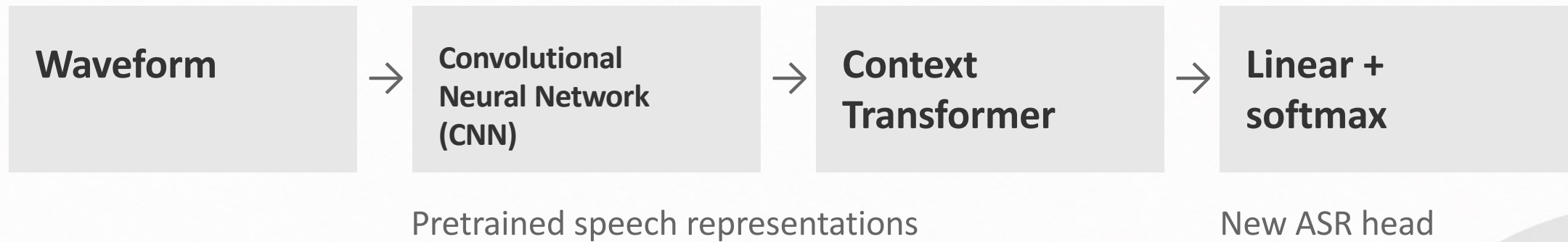
ASR learns an alignment

Many acoustic frames correspond to fewer text tokens



wav2vec 2.0 becomes an ASR model

Fine-tune with Connectionist Temporal Classification (CTC)



Each frame → probabilities over letters + blank

Training: CTC loss

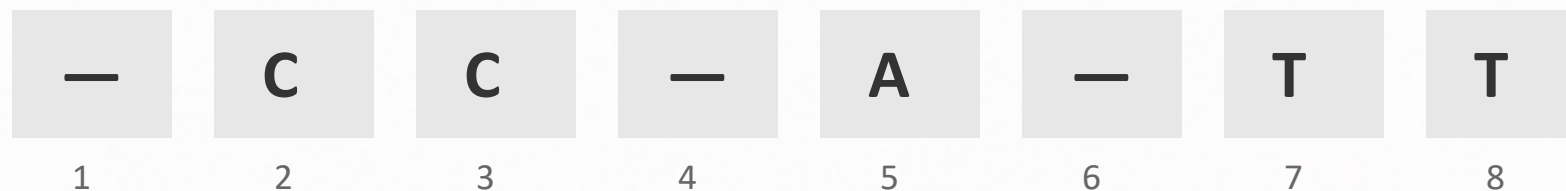
Inference: decode frame labels

CTC turns frame labels into text

“—” = blank: emit no character at this frame.

It is not a word boundary or a silence label.

Frame labels



1 Merge repeats

— C — A — T

2 Remove blanks

CAT

Blank separates repeated letters: $LL \rightarrow L$ $L - L \rightarrow LL$

CTC loss adds up the correct paths

A two-frame teaching example · target transcript: A

Symbol	Frame 1	Frame 2
A	0.6	0.7
blank	0.4	0.3

$$A A \rightarrow A \quad 0.6 \times 0.7 = 0.42$$

$$A - \rightarrow A \quad 0.6 \times 0.3 = 0.18$$

$$- A \rightarrow A \quad 0.4 \times 0.7 = 0.28$$

— — gives empty text, so it is excluded.

$$\mathcal{L}_{\text{CTC}} = -\log \sum_{\pi: B(\pi)=y} \prod_{t=1}^T p_t(\pi_t | x)$$

$$P(A | x) = 0.88 \quad \mathcal{L}_{\text{CTC}} = -\ln(0.88) \approx 0.128$$

What makes a voice recognizable?

From content representations to speaker embeddings

A speaker embedding

A recording becomes a fixed-length vector.



Speech recording

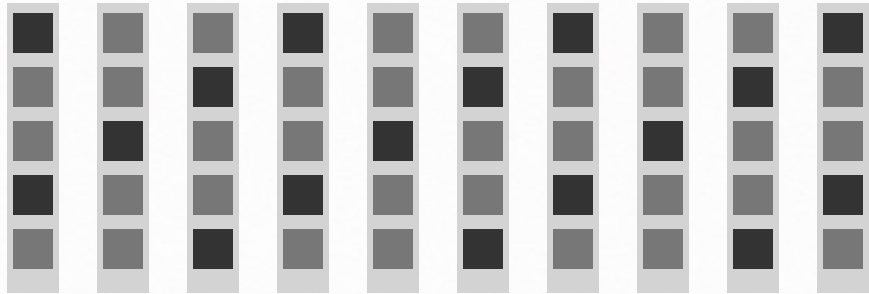
Variable-duration input

Embedding e

Fixed-length representation

Pooling collects evidence over time

Frame features become one utterance representation.



Frame-level features

Pooling



One vector

Speaker embeddings with contrastive learning

Supervised contrastive learning uses speaker identifier (ID) labels.

Same speaker

Different clips form a positive pair.

Pull their embeddings closer.

Different speakers

Clips form a negative pair.

Push their embeddings apart.

Use the embedding for speaker recognition

Verification: same speaker?

Identification: which enrolled speaker?

How speaker representations evolved

Different models, the same fixed-vector goal

i-vector

Statistical utterance representation

d-vector

Aggregate neural speaker features

x-vector

Temporal network + statistics pooling

Speaker encoders used today

Representative models and tools

ECAPA-TDNN

Emphasized Channel Attention, Propagation and Aggregation (ECAPA)
Time Delay Neural Network (TDNN)

Resemblyzer

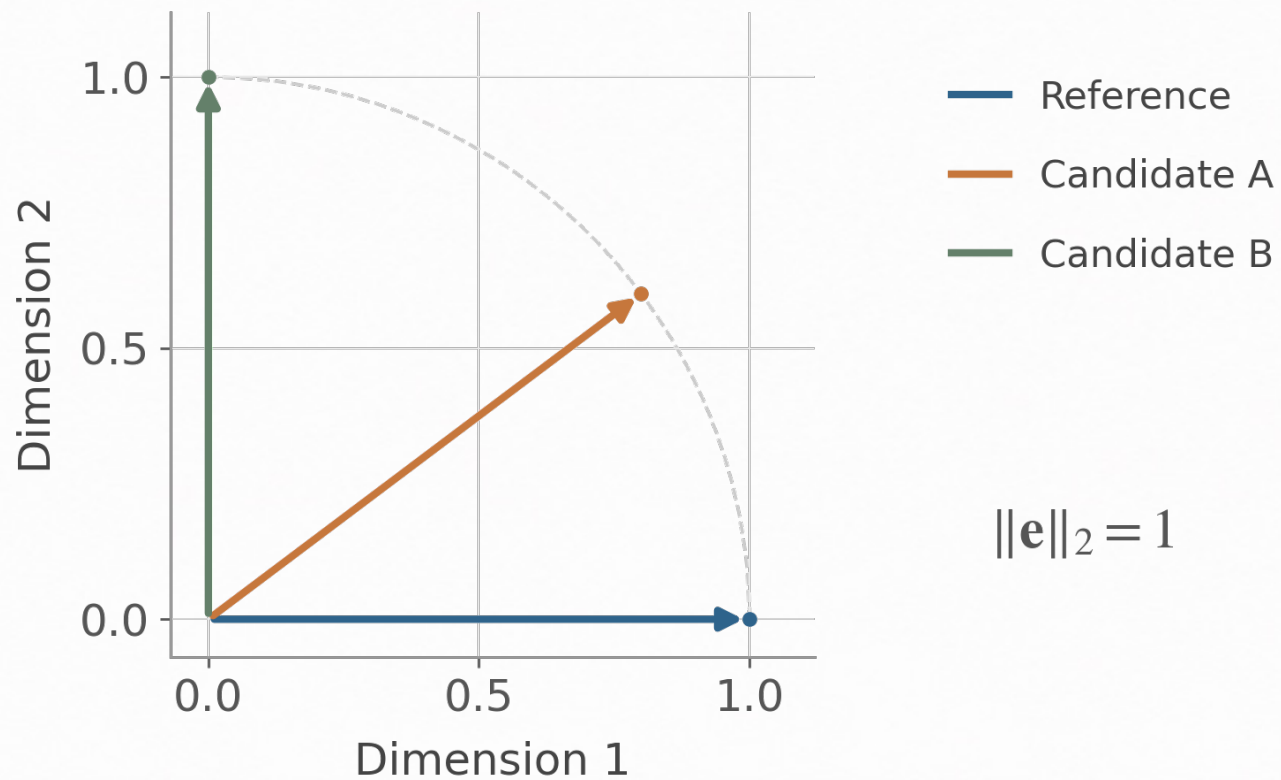
Generalized End-to-End (GE2E) training
256-dimensional speaker embeddings

WavLM + head

Self-supervised front end, speaker training

Similarity is geometry

Illustrative two-dimensional unit vectors



Small angle
High cosine

Cosine similarity

The angle matters after normalization.

$$\text{sim}(\mathbf{e}_1, \mathbf{e}_2) = \frac{\mathbf{e}_1 \cdot \mathbf{e}_2}{\|\mathbf{e}_1\|_2 \|\mathbf{e}_2\|_2}$$

Who spoke when?

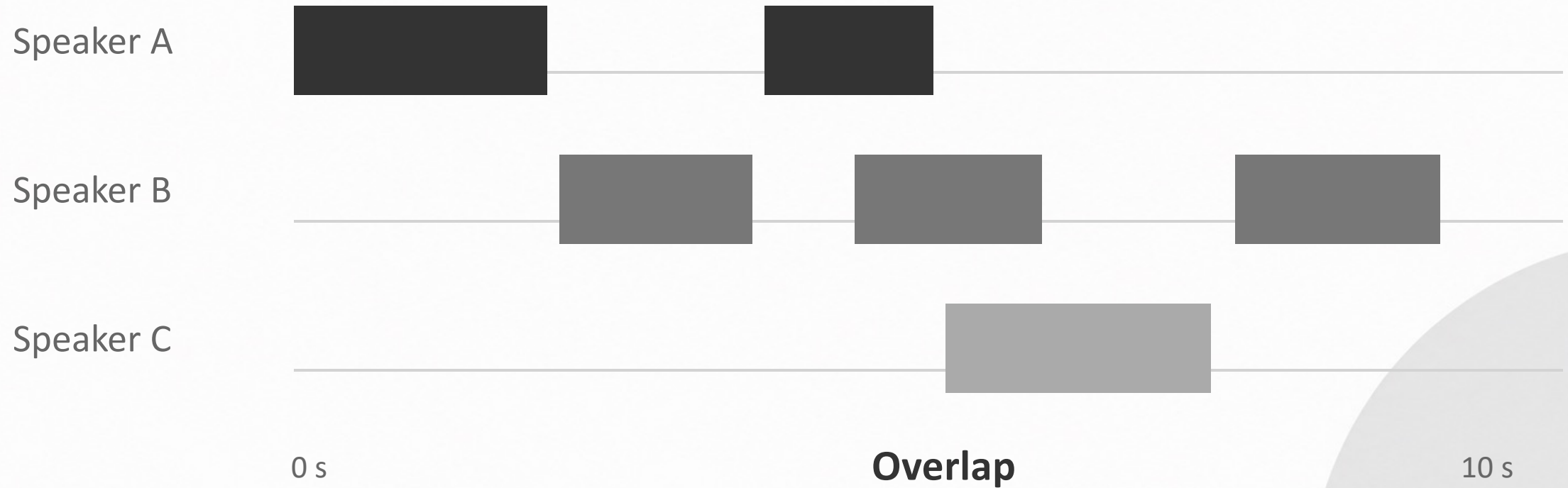
Anonymous speaker labels over time



10 s

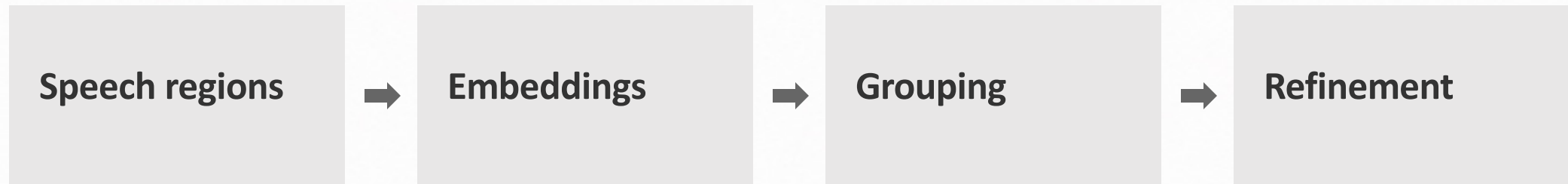
People overlap

Two voices can occupy the same moment.



A diarization pipeline

Detection, representation, grouping, refinement



Words and speaker turns are complementary

The same representation can support different decisions.

ASR

What was said?

Speaker diarization

Who spoke when?

What happened beyond the words?

Speech and audio understanding

Understanding needs audible evidence

Words

The content actually spoken.

Sound events

What happened, and when:
a cup clink, traffic, a door.

Music

Instruments, melody, harmony, rhythm and musical style.

Voices

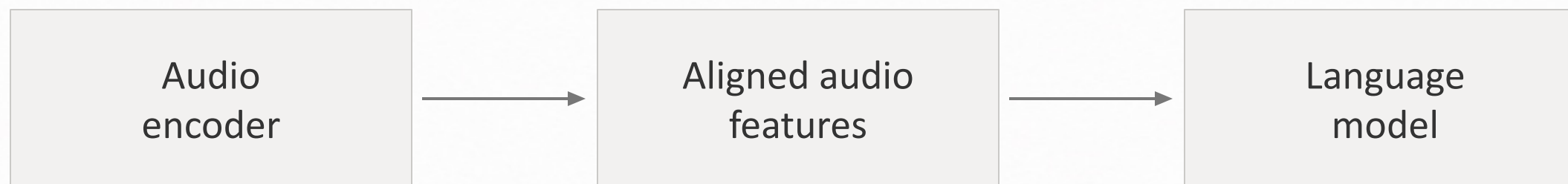
Each speaker, and their delivery:
pitch, stress, rhythm, emotion.

Room acoustics

Reverberation, distance,
and the sense of space.

How an audio-language model listens

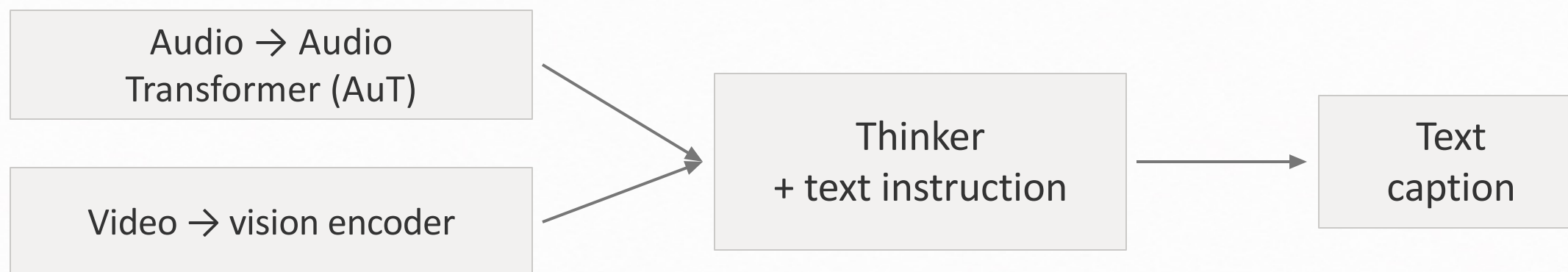
Audio features become context for a language model



Projection, resampling or cross-attention can connect the two.
A separate adapter block is not required.

Qwen3-Omni: from audio to a caption

Audio, video and text meet in the Thinker



Time-aligned Multimodal Rotary Position Embedding (TM-RoPE)
aligns audio and video in time.

OFFICIAL DEMO · CAPTION SHORTENED

Rhythmic jackhammer impacts over a low machinery hum;
a high-pitched cutting sound joins later.

The frontier: longer, richer audio context

Qwen3.5-Omni

Long audio context

Track events and speaker references over time

Audio-visual grounding

Connect audible and visible evidence

Structured captions

Describe events with temporal organization

Did the model really listen?

Many benchmark questions can be answered from text or vision alone.
A high score can hide weak use of acoustic evidence.

“Does the voice sound female or male?”

ACTUAL AUDIO

Female-sounding voice

“I’m a little boy.”

POSSIBLE TEXT-DRIVEN ANSWER

“Male.”

Follows “boy”, overlooks the voice.

**Captions can miss acoustic details.
Reliable audio understanding remains an open problem.**

So, how should we evaluate audio?

First decide what we want the system to get right.

What exactly are we evaluating?

Content accuracy

Are the intended words preserved?

Naturalness

Does the delivery sound natural?

Audio quality

Is the signal free of noise and artifacts?

Speaker similarity

Voice conversion: does it match the target voice?

Acoustic similarity

Does the room, reverberation or ambience match?

Mean Opinion Score (MOS)

Mean listener rating; the question defines the scale.

Example rating question	1	5
Audio quality	Bad	Excellent
Speaker similarity	Completely different voice	Perceptually the same voice

MOS · one item, often 1–5; no exact clean version required.

Multiple Stimuli with Hidden Reference and Anchor (MUSHRA)

Same-content versions, 0–100, with a reference and anchors.

Your turn to rate

Overall listening quality: 1 bad ... 5 excellent



A

3.9 s



B

3.9 s

Listen first. Then compare.

Low-quality anchors can check attention and calibrate the scale.

Exclude listeners only by predeclared quality-control rules.

A listening test is an experiment

- 1 Define the question**
- 2 Collect human responses**
Blind conditions and randomize the presentation order.
- 3 Filter out invalid responses**
- 4 Report results and uncertainty**

Human MOS and predicted MOS

A model learns to predict human quality ratings.

Deep Noise Suppression
Mean Opinion Score (DNSMOS)

P.835: speech signal, background noise,
and overall quality.

Non-Intrusive Speech
Quality Assessment (NISQA)

Overall quality plus noisiness,
coloration, discontinuity and loudness.

Useful for screening; validate important comparisons with listeners.

Now improve the recording

Quality evaluation gives enhancement a target

Speech enhancement

Noisy observation, cleaner estimate

$$y(t) = s(t) + n(t)$$

y	s	n
observation	speech	noise

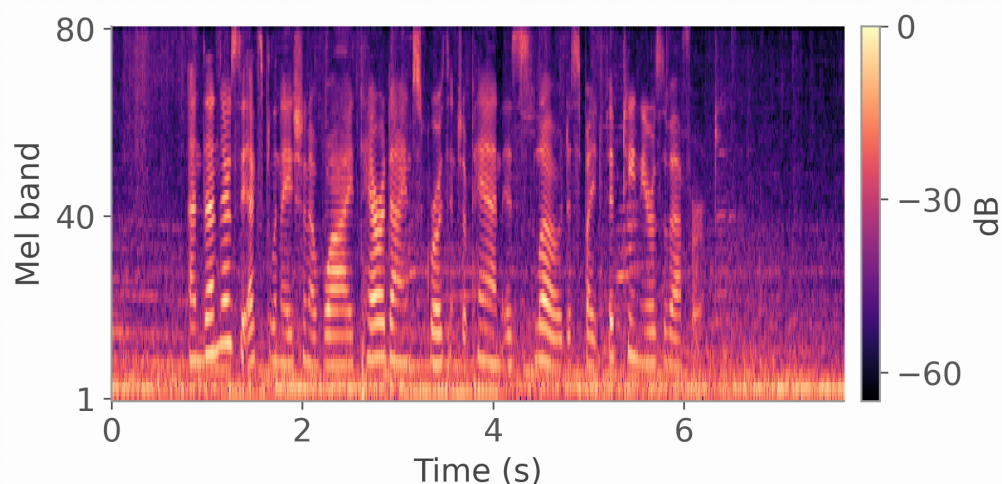
Recover speech while preserving words and voice.

Enhancement: suppression and generation

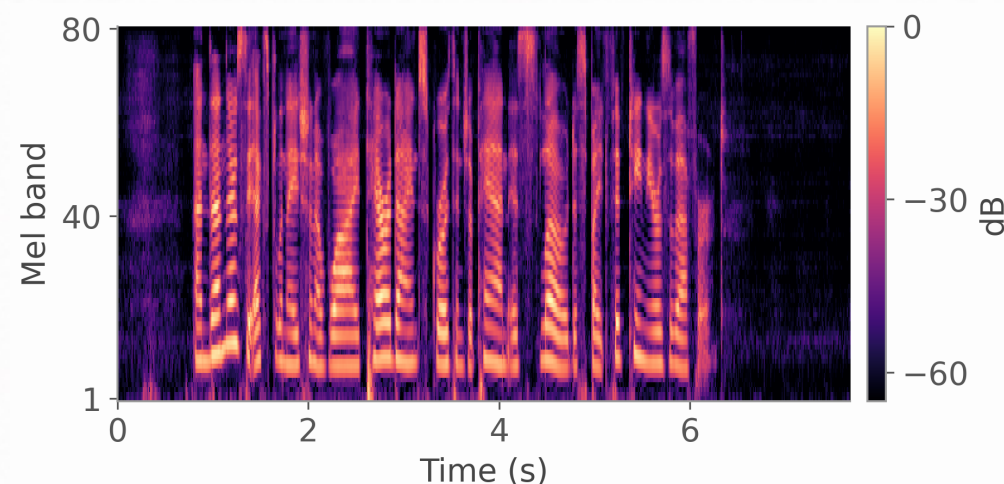
Score-based Generative Models for Speech Enhancement (SGMSE)

The example below uses the SGMSE+ diffusion enhancer.

Noisy input



SGMSE+ · diffusion output



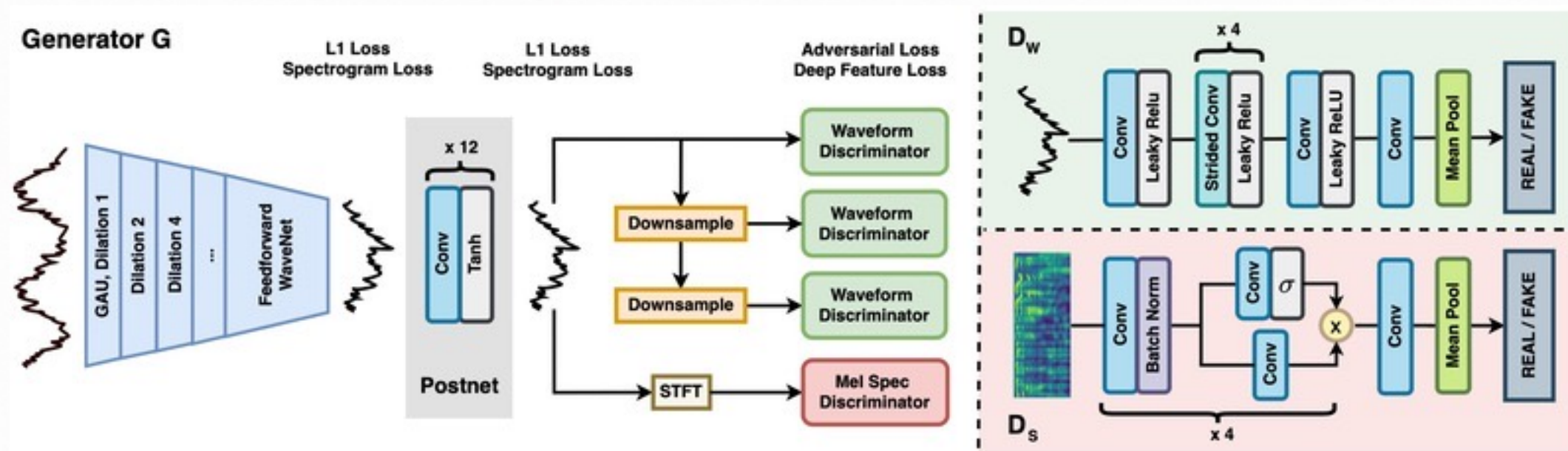
Aggressive masks can remove speech detail.

Generation can restore detail, but must preserve what was said.

HiFi-GAN for speech enhancement

Jiaqi Su, Zeyu Jin and Adam Finkelstein · Interspeech 2020

Early research behind Adobe Enhance Speech



Generative Adversarial Network (GAN): waveform + Mel discriminators

Adobe Podcast Enhance Speech



Click video to play / pause

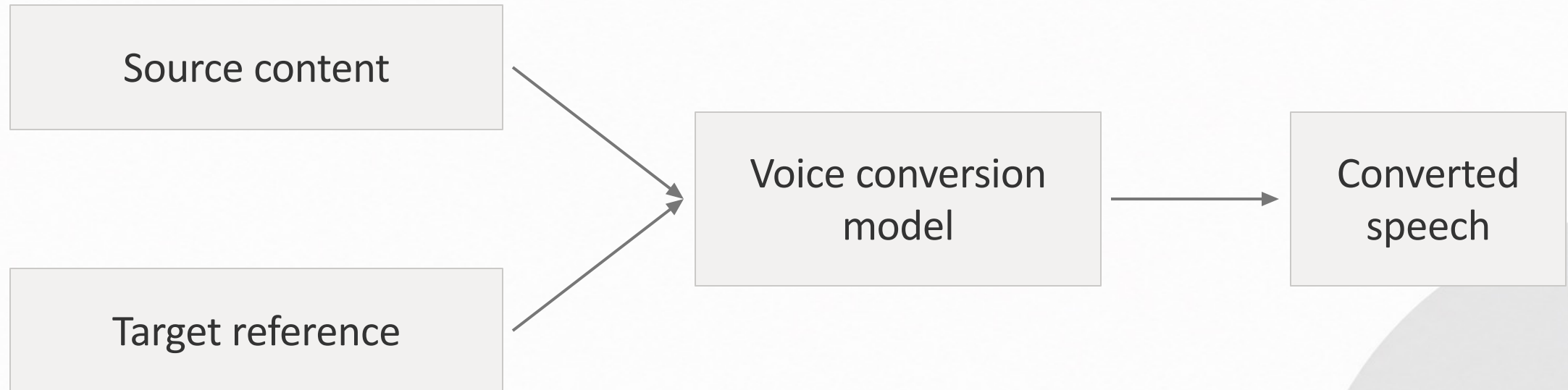
[Open video online ↗](#)

Now change the voice

Keep the words. Change the speaker's voice.

Voice conversion

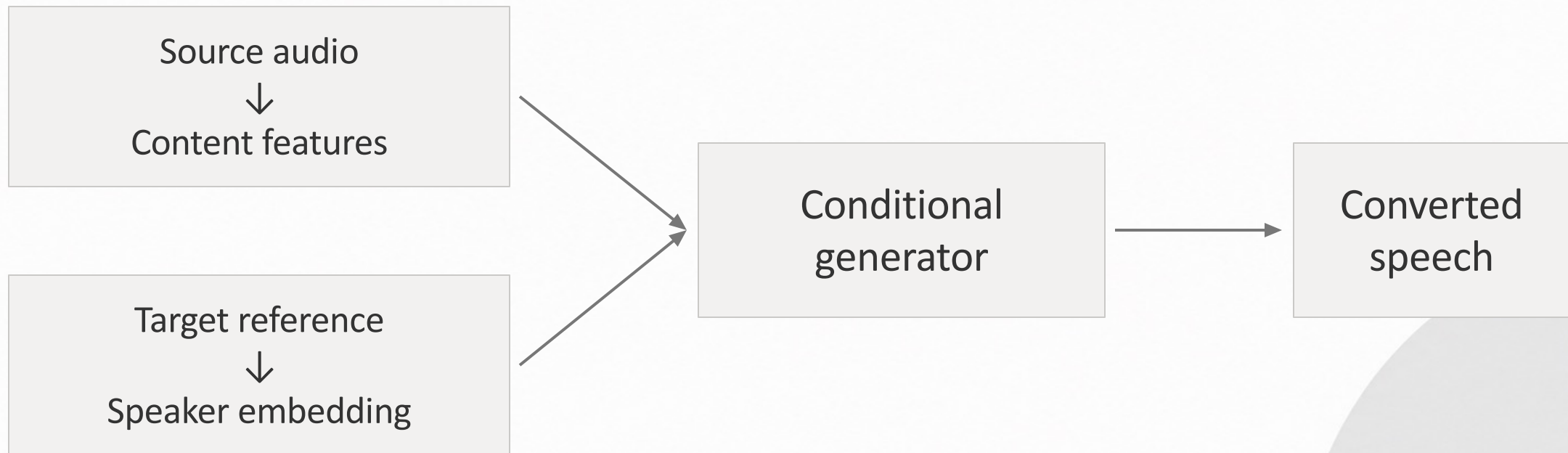
Source content + target voice reference



The reference provides the voice; its words can be different.

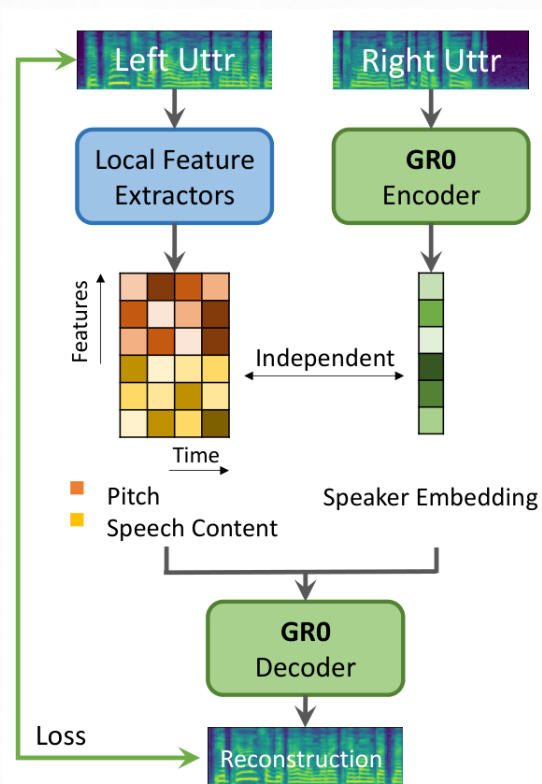
Speaker embedding can condition a voice

Content and speaker identity enter as separate conditions.



GR0: global and local representations

Self-supervised Global Representation Learning for Zero-shot Voice Conversion



Local: a sequence over time

Speech content and normalized pitch from the source segment.

Global: one vector for the whole clip

Voice characteristics shared across different content from the same speaker.

Learn the global vector through reconstruction

Local features from one segment + global vector from another segment of the same speaker.

Source. Reference. Converted

Female source, male reference, different sentences.

01

Source



4.3 s

02

Reference



3.1 s

03

Converted

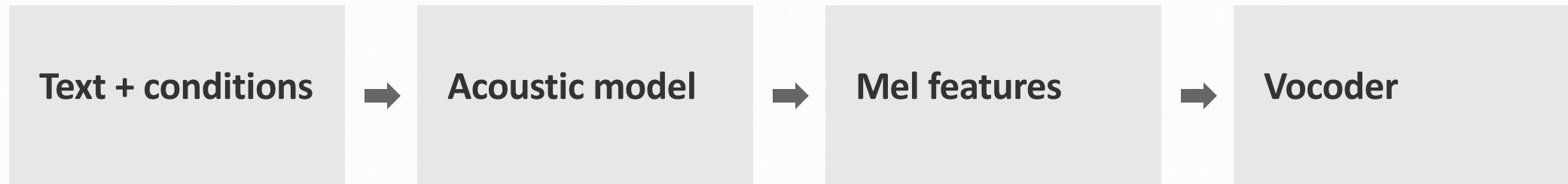


4.3 s

Judge content, speaker similarity and naturalness separately.

Text-to-speech (TTS) chooses an acoustic realization

The text specifies words, but leaves many choices open



Early neural text-to-speech

A short history of neural text-to-speech

Earlier TTS

Concatenative and statistical-parametric systems

Deep Voice · 2017

Neural phonemes, duration and pitch

Tacotron · 2017

Text → Mel → waveform reconstruction

Vocoder: Mel features become a waveform

Generative Adversarial Network (GAN) vocoders render audio.

HiFi-GAN

GAN vocoder; V1 / V2 / V3 configurations

BigVGAN

Periodic activations + anti-aliasing

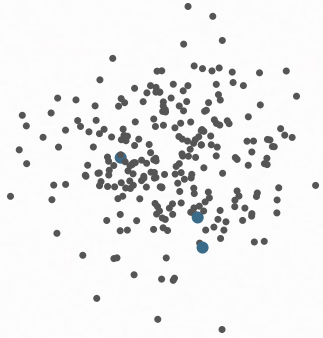
BigVGAN-v2

Broader audio training; updated objectives

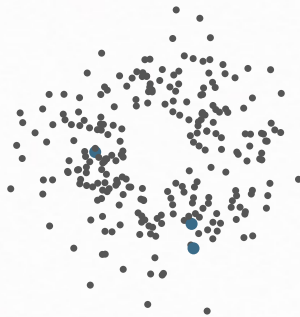
Diffusion and flow matching

2D toy example: Gaussian noise becomes a ring-shaped distribution.

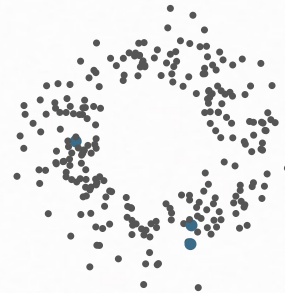
Gaussian noise



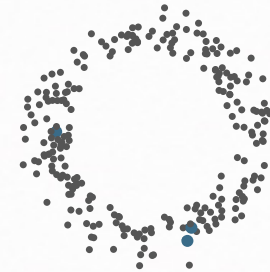
$t = 0.25$



$t = 0.50$



$t = 0.75$



Ring data



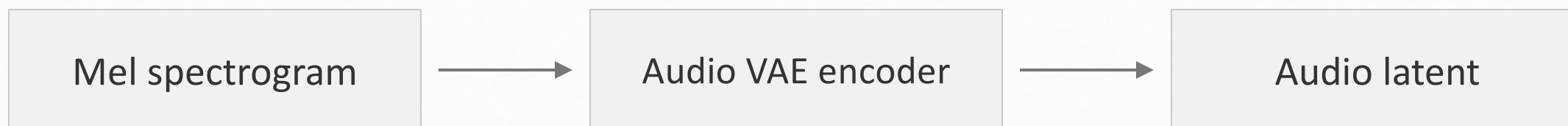
Diffusion: learn to denoise at each noise level.

Flow matching: learn a velocity field for the updates.

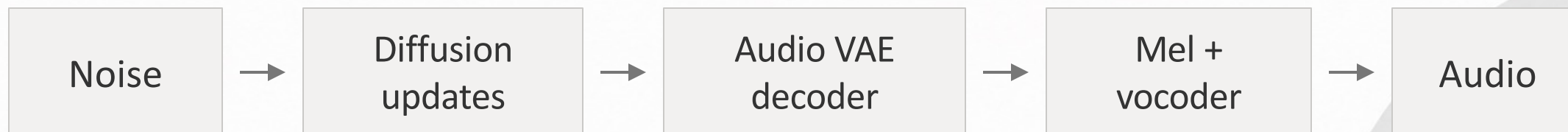
Audio latent diffusion

A Variational Autoencoder (VAE) provides a compact audio latent space.

EXAMPLE: AUDIOLDM



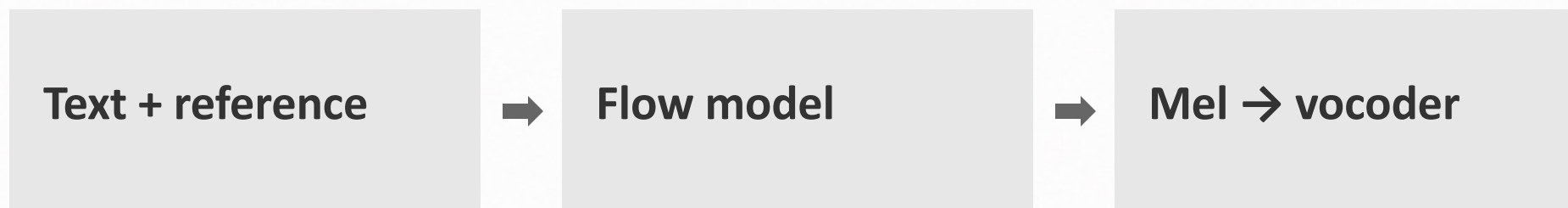
GENERATION · CONDITIONED ON A TEXT PROMPT



F5-TTS instead applies flow matching directly to Mel spectrograms.

F5-TTS connects flow matching to speech

Text + reference audio → generated Mel → waveform



Better generation needs several checks

Quality gains are empirical—not guaranteed by model family

01

Quality

MOS / preference

02

Content

Words preserved

03

Voice

Reference similarity

04

Control

Instruction followed

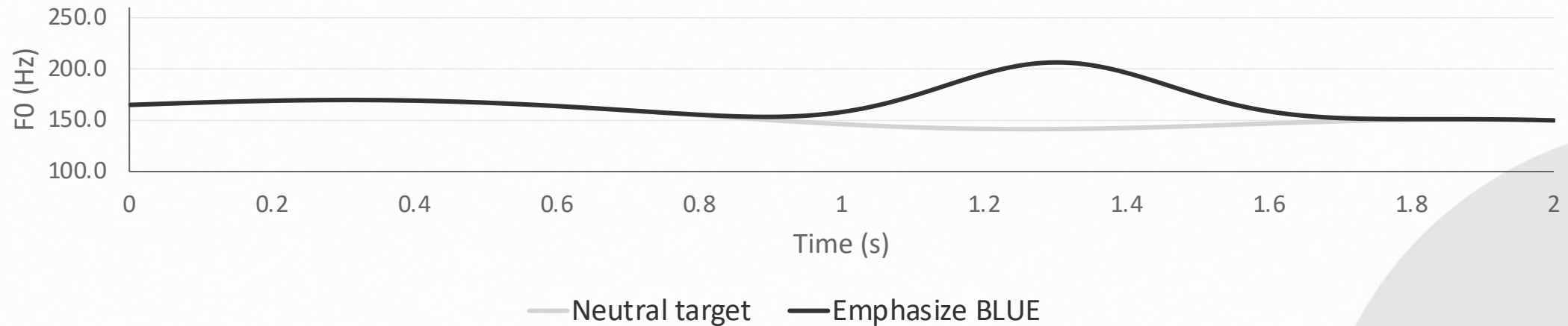
Can we direct it like a voice actor?

Future directions: precise control over speech

Pitch and emphasis need a target

“I wanted the BLUE one.”

I wanted the BLUE one.



Illustrative voiced-region targets

Timing control lives on several scales

A pause, a word, a phoneme—and the whole utterance

0.00 s

2.00 s



A longer local segment must fit the global time budget.

Even “th” is not one sound

Pronunciation control must reach below the word

thin /θ/

Voiceless dental fricative

this /ð/

Voiced dental fricative

Identity / duration / realization

A direction should be executable

Example of a desired interface—not a claimed model capability

Voice	Keep the reference speaker
Word	Emphasize BLUE
Contour	Follow the local pitch curve
Timing	Pause 200 ms; finish at 2.0 s
Phoneme	Edit one onset; preserve the rest

How would you evaluate controllability?

Compare the request with the realized recording

Adherence

Pitch trajectory, boundaries, duration budget

Perception

Did listeners hear the intended emphasis?

Locality

Did unrelated words and identity stay stable?

Q&A